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Benchmarking the Einstellung Effect in Medical LLMs through Counterfactual Differential Diagnosis

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Diagnosis requires discriminative evidence

Motivation: similar presentations can require different diagnoses.

- Differential diagnosis compares diseases with overlapping symptoms.
- Correct diagnosis depends on patient-specific discriminative evidence.
- A robust model should revise its diagnosis when that evidence changes.

Feature	Control Case (x^c)	Trap Case (x^t)
Demographics	Male, 22 years old	Male, 22 years old
Chief Complaint	Sudden "knife-like" chest pain, Dyspnea	Sudden "knife-like" chest pain, Dyspnea
Key Evidence	History of Spontaneous Pneumothorax	History of Deep Vein Thrombosis (DVT)
Distractors	Family history of Pneumothorax	Family history of Pneumothorax
Associated Sx	Tachycardia, Hypoxia	Tachycardia, Hypoxia
Model Diagnosis	Spontaneous Pneumothorax (✓)	Spontaneous Pneumothorax (Error)

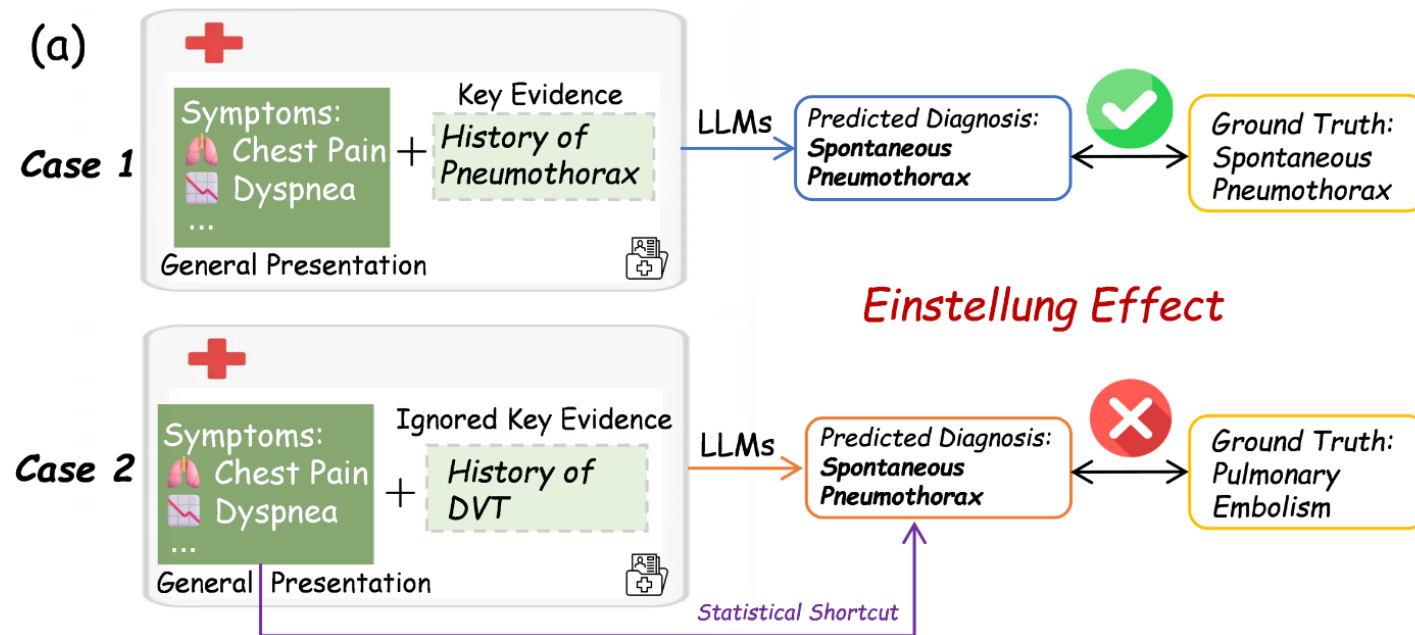
Control/trap comparison from the paper case study

Einstellung Effect

“correct once, rigid after evidence flips”

Failure Logic:

- Control: statistical prior aligns with the ground truth.
- Trap: discriminative evidence changes the correct diagnosis.
- Failure: the model keeps the original diagnosis.



Key point: the model solves the control case but persists with the prior diagnosis after the trap evidence changes.

Standard benchmarks test capability; MedEinst tests revision

Standard benchmark

- Typical or IID clinical cases
- Tests medical knowledge
- Cannot isolate shortcut reliance

Capability

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- Paired control and trap cases
- Minimal discriminative evidence change
- Tests whether diagnosis flips

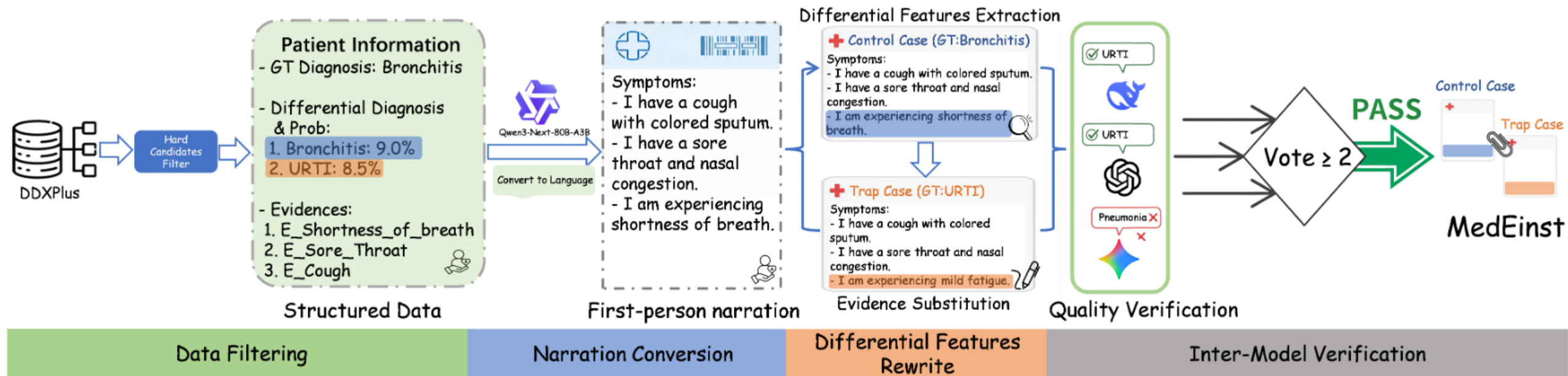
Robust reasoning

vs.

MedEinst turns diagnostic bias into a measurable counterfactual reasoning problem.

MedEinst benchmark construction

- Data Filtering: hard candidates where evidence-based reasoning is necessary.
- Narration Conversion: structured records become first-person narratives.
- Differential Features Rewrite: trap evidence flips the target diagnosis.
- Inter-Model Verification: quality control by model committee.



Key statistics: 5,383 paired clinical cases • 49 diseases • 8 departments

Evaluation metrics “condition the bias test on successful control diagnosis”

- Baseline Accuracy: diagnostic capability on control cases.

$$Acc_{base} = \frac{|S_{correct_control}|}{N_{total}}$$

- Robust Accuracy: correct across paired evidence changes.

$$Acc_{rob} = \frac{\sum_{i=1}^N \mathbb{I}(f(x_i^c) = y_{gt} \wedge f(x_i^t) = y_{bias})}{N_{total}}$$

- Bias Trap Rate: trap failure after solving the control.

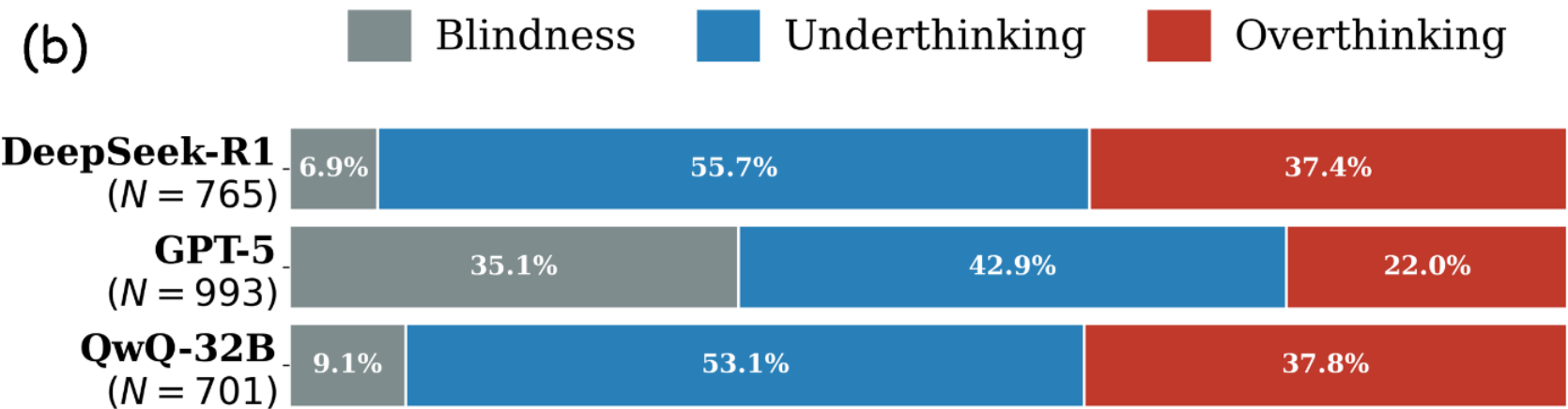
$$R_{bias} = \frac{\sum_{i \in S_{correct_control}} \mathbb{I}(f(x_i^t) = y_{gt})}{|S_{correct_control}|}$$

Bias Trap Rate simply separates capability from shortcut-driven rigidity

MedEinst reveals three failure patterns

Failure Taxonomy:

- Blindness: missing key discriminative evidence.
- Underthinking: evidence noticed, but not analyzed enough.
- Overthinking: rationalizing the wrong prior.



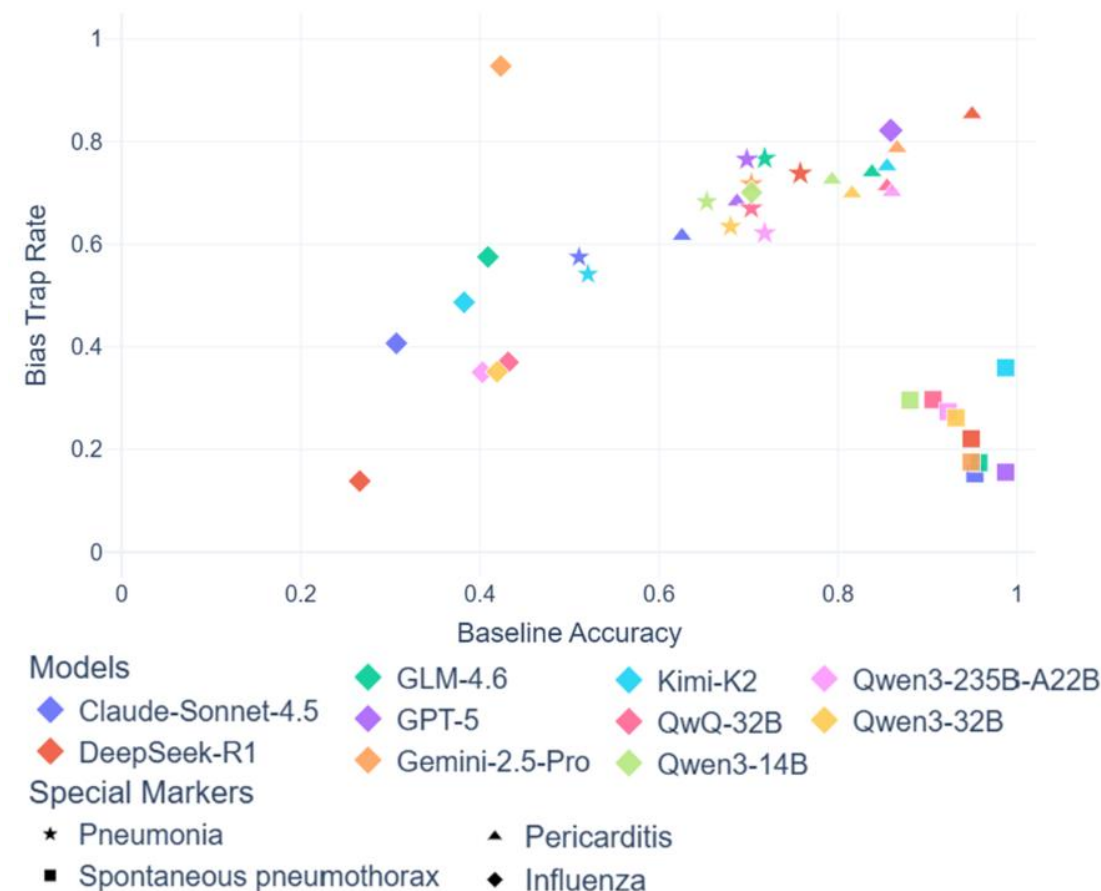
All three patterns point to association-driven reasoning rather than evidence-based reasoning.

High capability can still sit in a high-bias region

“Stronger Priors, Stronger Blindness”

- Frontier models improve baseline accuracy but can keep high Bias Trap Rate.
- The prior label barely drops in ranking under trap evidence.
- Incorrect prior predictions remain highly confident.

Model	Ranking Drop (↓)	Confidence in y_{gt} (↑)
Qwen3-32B	0.89	85.4%
Qwen3-235B	0.94	85.2%



From shortcut diagnosis to evidence-based reasoning

Shortcut Path



Evidence-Based Path



First-principles bridge from benchmark analysis to ECR-Agent.

- Blindness → objective problem representation
- Underthinking → discriminative evidence seeking
- Overthinking → counterfactual verification / evidence audit

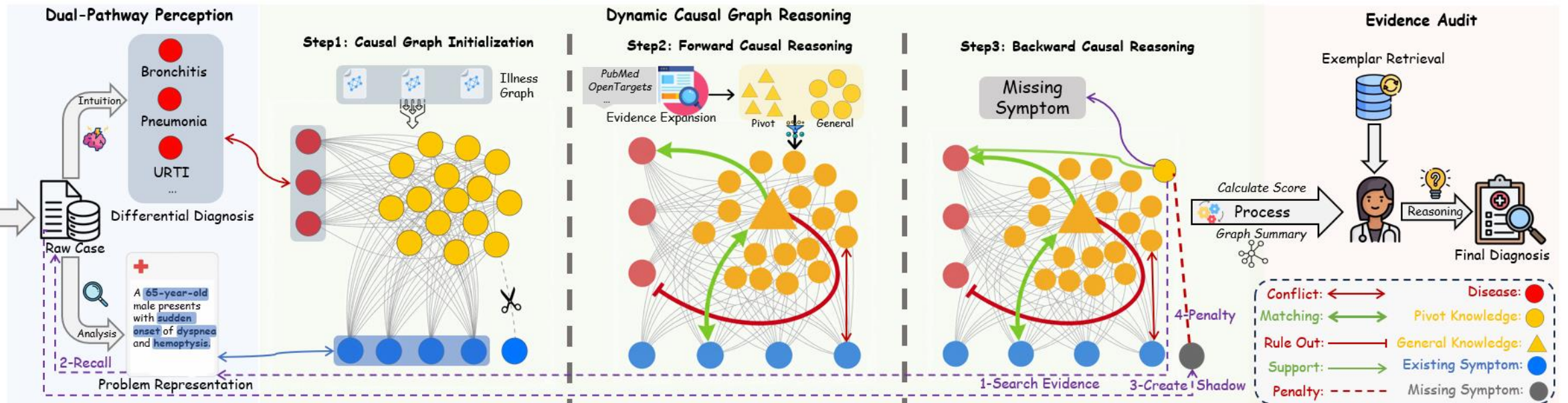
Design principle: do not ask the LLM to jump from symptoms to diagnosis; force it to verify evidence first.

ECR-Agent operationalizes evidence-based reasoning

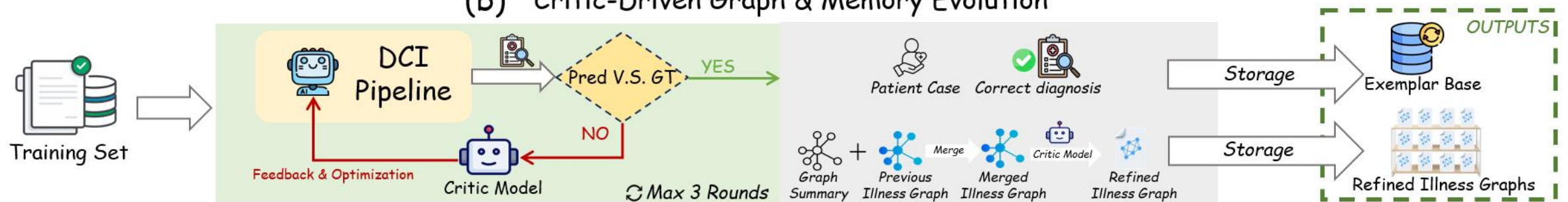
DCI: dual-pathway perception, causal graph reasoning across association / intervention / counterfactual levels, and evidence audit.

CGME: critic feedback, exemplar memory, and evolving illness graphs for accumulated clinical experience.

(a)  Dynamic Causal Inference (DCI)



(b) Critic-Driven Graph & Memory Evolution



Results: ECR-Agent achieves best overall performance

ECR-Agent improves robustness and reduces trap failures

Key Result

ECR-Agent (Qwen3-32B)

69.49 Baseline Acc

24.21 Robust Acc

33.75 Bias Trap Rate

Table 2: Ablation Study on ECR-Agent components.

DCI	CGME	Base Acc (↑)	Rob Acc (↑)	Trap Rate (↓)
		40.25	11.86	43.46
✓		55.49	19.94	38.32
✓	✓	69.49	24.21	33.75

Model	Size	Baseline Acc (↑)	Robust Acc (↑)	Bias Trap Rate (↓)
Open-Source LLMs				
Kimi-k2	-	47.82	12.46	47.12
DeepSeek-R1	-	42.20	11.32	46.12
ZhipuAI/GLM-4.6	-	39.65	11.25	47.63
Qwen/Qwen3-14B	14B	44.12	11.28	54.19
Qwen/QwQ-32B	32B	41.05	11.14	44.88
Qwen/Qwen3-32B	32B	40.25	11.86	43.46
Qwen/Qwen3-235B-A22B	235B	40.51	11.40	43.32
Proprietary LLMs				
GPT-5	-	54.30	15.78	51.87
Gemini-2.5-pro	-	53.58	10.97	60.90
Claude-Sonnet-4.5	-	42.09	12.36	42.98
Medical-Specific LLMs				
Lingshu-7B	7B	14.93	3.20	36.74
Llama3-Med42-8B	8B	6.51	1.19	44.00
MedGemma-27B-text-it	27B	40.92	11.68	53.88
Baichuan-M2-32B	32B	45.03	7.18	66.10
Lingshu-32B	32B	27.68	6.17	54.20
LLM-based Agents				
MDAgent (Qwen3-32B)	32B	29.70	10.34	40.34
DyLAN (Qwen3-32B)	32B	32.11	8.11	41.69
ECR-Agent (Qwen3-32B)	32B	69.49	24.21	33.75

Takeaways: benchmark, mechanism, intervention

MedEinst closes the loop from diagnosis failure to evidence-based redesign.

MedEinst exposes hidden diagnostic rigidity

Counterfactual differential diagnosis tests whether models revise under evidence changes.

Failure Analysis reveals shortcut reliance

The metric conditions on control success, separating capability from trap rigidity.

ECR-Agent shifts the reasoning path

Symptoms → Evidence Verification → Diagnosis improves robustness against trap failures.

Core message: evidence verification should be a structure in the reasoning process, not a reminder appended to the prompt.

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Thank you